



Department of Computer Science and Engineering

Academic Year 2021 – 2022(Odd / Even Semester)

Degree, Semester & Branch: B.E, I & CIVIL

Course Code & Title: GE3151 & Problem Solving and Python Programming

Name of the Faculty member (s): Dr.M.Gomathy Nayagam

Innovative Practice Description

- **Unit / Topic:** Unit I / Illustrative Problems (Towers of Hanoi)
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 1,2,3
- **Activity Chosen:** Game based Learning
- **Justification:**

The Tower of Hanoi is a mathematical game or puzzle consisting of three rods and a number of disks of various diameters, which can slide onto any rod. This active learning strategy is chosen to help the students to easily understand the concept of towers of Hanoi problem by playing game.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

The course instructor shared the online gaming link to the students in the class. The following is the link for online towers of Hanoi game.

https://www.mathplayground.com/logic_tower_of_hanoi.html

The course instructor had chosen the volunteers to play the game in front of the class. Also, he explained the rules for playing to the students. Mr.Janarathanan of the class come front as volunteer to play the game. He started the game and explained about the game to class as shown in fig 1.a,b. Finally, the course instructor explained to write the algorithm for game.



(a)



(b)

Fig.1. Mr. Janarathanan is Playing Towers of Hanoi Game in online

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 5	PO 10	PO12	PSO 3
CO 1	3	2	2	1	2	1	1

(1 – Low

2 – Moderate

3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 1	PO 2	PO 3	PO 5	PO 12
	3	2	1	1	2
Justification for correlation	Students could be able to apply basic engineering theory and principles to solve the towers of hanoi problem	Students could be identify their own example to explain the algorithm	Students could be able to recollect the basic principles of towers of hanoi algorithm	Students could able choose the modern tool to interpret and explain the algorithm for towers of hanoi	Students interacted and shared their views about the concept with their peer group by ethically

Innovative practice	PSO 3
	2
Justification for correlation	Students could be able to provide the solution for real time problems.

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

Students felt that this activity is very interesting to play, useful and got good idea about the development of algorithm for towers of hanoi

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Most of the students understand and write the step by step procedure of guessing an integer number individually.
- Student's interaction with their peer members and the class during the activity was good and effective.

❖ **Challenges faced in implementation:**

- Due to time limitation, only few students were made to do the activity.

References:

- ❖ https://www.mathplayground.com/logic_tower_of_hanoi.html
- ❖ [https://edtechreview.in/dictionary/298-what-is-game-based-learning#:~:text=Game%20based%20learning%20\(GBL\)%20is,matter%20to%20the%20real%20world.](https://edtechreview.in/dictionary/298-what-is-game-based-learning#:~:text=Game%20based%20learning%20(GBL)%20is,matter%20to%20the%20real%20world.)
- ❖ [https://tophat.com/glossary/g/game-based-learning/.](https://tophat.com/glossary/g/game-based-learning/)

Signature of Faculty Member

HOD